

Prathibha Kalagani

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EDUCATION

New York Institute of Technology, Master of Science, Computer Science

New York, NY, USA | Jan 2024 – May 2025

International Institute of Information Technology (IIIT), Bachelor's in Computer Science

Srikakulam, India | Aug 2018 – Jun 2022

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript, TypeScript

Front-end: HTML, CSS, JavaScript, React, Angular, Next.js, GraphQL, Redux.

Backend: Spring Boot, Node.js, Kafka, GraphQL, REST APIs, Postman

Databases: MySQL, PostgreSQL, Amazon DynamoDB, MongoDB, MS SQL Server, Couchbase

Cloud & Ops: AWS (ECS, Lambda, SQS, CloudWatch, S3), Google Cloud, Docker, Kubernetes, Terraform, Jenkins, CI/CD, Git

Tools & Environments: GitHub, Maven, IntelliJ IDEA, VS Code, Android Studio, UNIX

WORK EXPERIENCE

US BANK, Full Stack Developer

May 2025 – Present

- Built end-to-end features by integrating **React.js with Spring Boot REST APIs**, ensuring smooth data flow, consistent state management, and secure UI-backend communication.
- Architected backend modules using **Java Spring Boot, implementing Controller Service Repository layers**, DTO mappings, validation filters, and centralized exception handling for predictable API behavior.
- Developed **React dynamic UI components** for form-driven workflows with schema-based validation, conditional rendering, multi-step logic, and debounced API lookups for enhanced performance.
- Utilized **Hibernate** to design entity models, configure one-to-one and one-to-many relationships, optimize fetch strategies, and execute high-performance queries for **large financial datasets**.
- Refactored legacy React class components into **modular functional components**, applying lazy loading, Suspense, and dynamic imports to reduce bundle size and improve runtime efficiency.
- Built automated test suites using **JUnit, Mockito, Postman Collections, and React Testing Library**, strengthening API contract reliability, reducing regressions, and improving release quality.

New York Institute of Technology, Graduate Teaching Assistant | *java, Python, DS, presentation skills*

Jan 2025 - May 2025

- Assisted students in mastering data structures, algorithms, and software engineering concepts, reinforcing problem-solving and code optimization skills.
- Developed automated grading scripts in Python to evaluate student submissions and identify performance bottlenecks, improving grading efficiency by 40%.

HCL Technologies Ltd, Full Stack Developer

Bangalore, India | Jun 2021 – Jan 2024

- Led development of **modular Spring Boot-based microservices** for a digital insurance modernization program, reducing inter-service coupling and enabling independent deployments, which cut release rollback risks by **30%**.
- Designed **event-driven policy update processing** using **Java, Spring MVC**, and asynchronous messaging improved claim adjudication response time under high-load conditions.
- Implemented **React Context API** for **state management**, allowing for more straightforward data sharing across the component tree without the need for **prop drilling**.
- Built **environment-agnostic CI/CD pipelines** in Jenkins with automated test execution, vulnerability scans, and canary release strategies, cutting deployment effort by **40%**, and reducing production incidents.
- Containerized microservices using Docker and **deployed on AWS ECS** with CI/CD integration and implemented **JUnit and Mockito** testing to enhance reliability and reduce defects.

IIIT Hyderabad, Software Engineering Research Intern

Hyderabad, India | Jan 2021 – Apr 2021

- Developed and tested **Spring Boot microservice prototypes** to analyze how architectural patterns affect service coupling and scalability, documenting findings that were later used to guide production of microservice decomposition.
- Built **React-based experimental UI components** to evaluate user interaction patterns in workflow-heavy insurance modules, improving form-flow validation accuracy, and reducing navigation friction in usability trials.
- Performed **database performance benchmarking** on PostgreSQL query plans and caching strategies, identifying indexing improvements that led to measurable reductions in lookup latency during research evaluations.

PROJECTS

Serverless E-Commerce Platform | *React, AWS DynamoDB, Cognito, Stripe, Lambda*

Jan 2024 – May 2024

- Designed and implemented a **serverless microservice architecture** using AWS Lambda, API Gateway, and DynamoDB, enabling event-driven product management, secure authentication through Amazon Cognito, and fault-tolerant order handling.
- Developed React components for product catalog, checkout and payment integration using Stripe and Amazon Cognito.